

BANSILAL RAMNATH AGARWAL CHARITABLE TRUST'S
VISHWAKARMA INSTITUTE OF TECHNOLOGY
(An Autonomous Institute affiliated to Savitribai Phule Pune University) PUNE
– 411037

Savitribai Phule Pune University



DEPARTMENT OF COMPUTER SCIENCE (SOFTWARE ENGINEERING)

Home assignment No.1

- Aim:**
1. Collect real-world categorical and numerical dataset & visualize distributions.
 2. Compare mean and median for skewed datasets and comment on which is better.
 3. Prepare a mini-report identifying best sampling strategy for a campus survey.

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1. Collect real-world categorical and numerical dataset & visualize distributions.

We used a **Student Performance Dataset** containing:

- **Categorical:** Gender, Pass
- **Numerical:** StudyHours, Attendance, Marks

Histogram (Numerical Distribution)

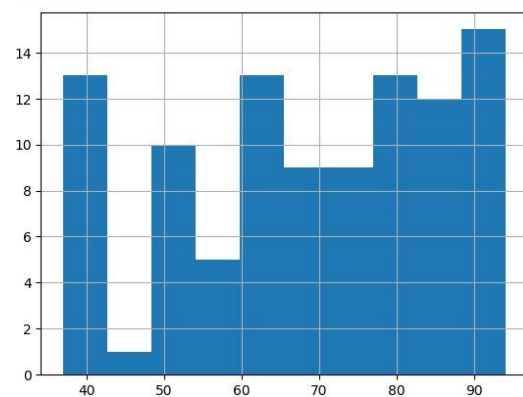
```
[16]: import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv('real.csv')
print(df)
```

	StudentID	Gender	StudyHours	Attendance	Marks	Pass
0	S001	Male	2	72	54	Yes
1	S002	Female	5	85	78	Yes
2	S003	Male	1	60	40	No
3	S004	Female	6	90	88	Yes
4	S005	Male	3	70	60	Yes
..	...	...	...	...	...	...
95	S096	Female	2	68	55	Yes
96	S097	Male	1	57	39	No
97	S098	Female	7	95	94	Yes
98	S099	Male	6	88	84	Yes
99	S100	Female	5	86	79	Yes

[100 rows x 6 columns]

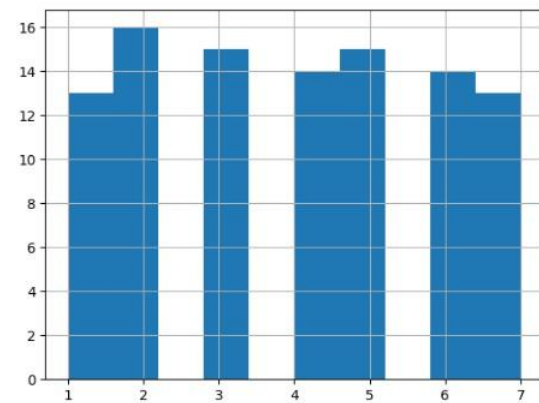
```
[17]: df["Marks"].hist()
```

[17]: <Axes: >



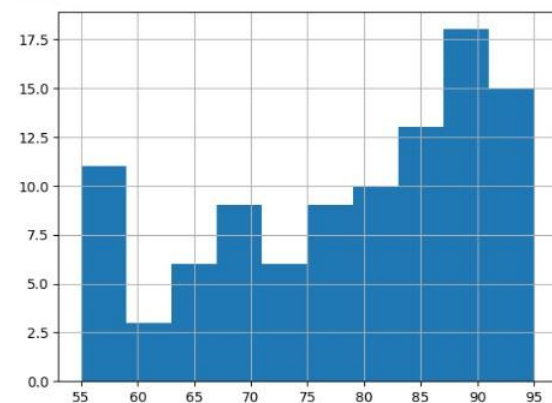
```
[18]: df["StudyHours"].hist()
```

[18]: <Axes: >



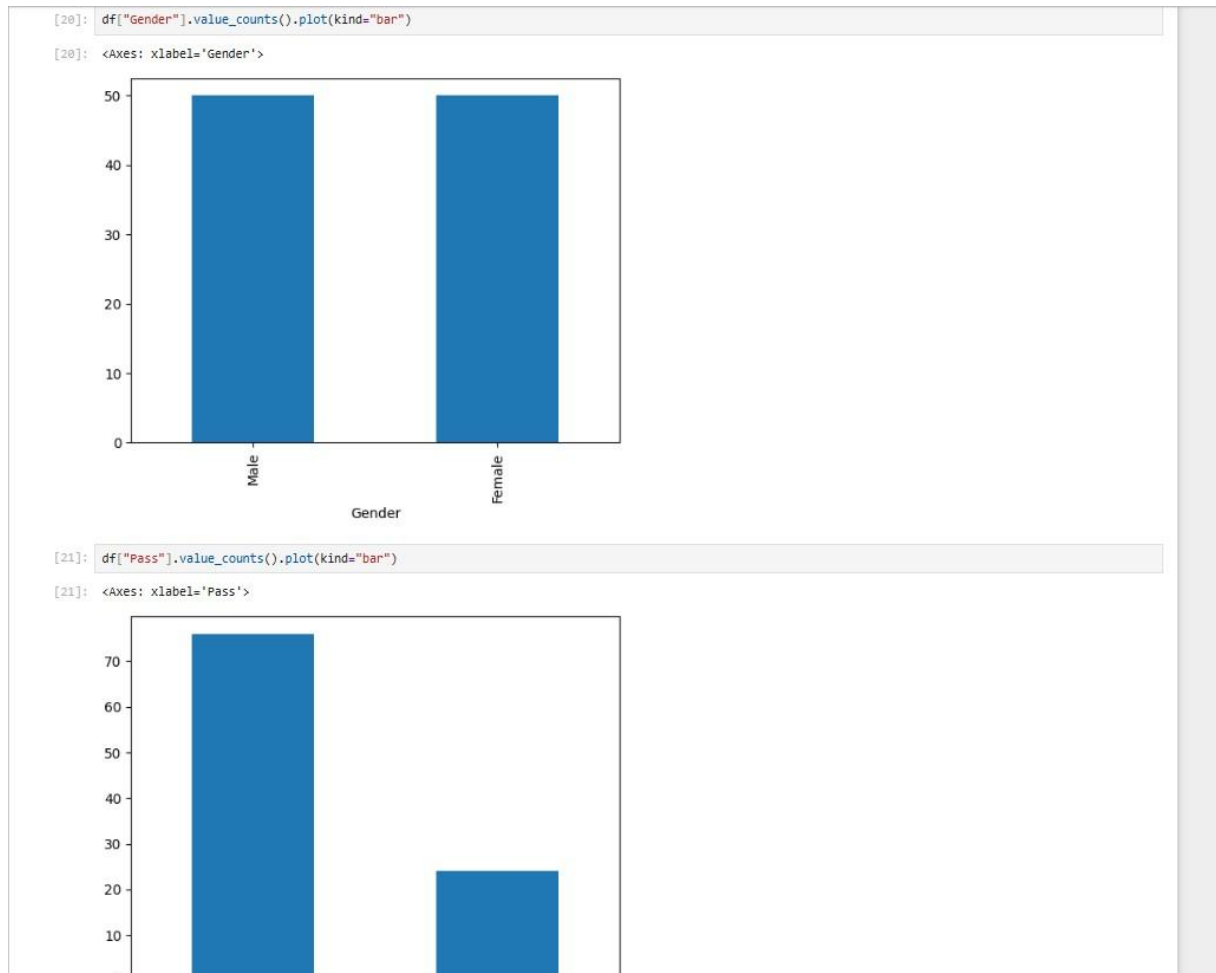
```
[19]: df["Attendance"].hist()
```

[19]: <Axes: >



Shows:

- Spread of values
- Skewness
- Frequency distribution



Shows:

- Category frequency
-

2. Mean vs Median (Skewed Data)

```
[23]: sum = 0
      count = 0
      for value in df["Marks"]:
          sum += value
          count += 1
      res = sum / count
      print(f"Mean = {res}")

Mean = 69.01

[29]: value = list(df['Marks'])
      value.sort()
      n = len(value)
      m = n // 2
      if m % 2 == 0:
          median = (value[m - 1] + value[m]) / 2
      else:
          median = (value[m]) / 2
      print(f"median = {median}")

median = 71.0

[]:
```

Normal Distribution

- Mean $\approx$ Median Both are good

Right-Skewed (e.g., Income, Sales)

- Mean > Median
Mean gets affected by large values

Left-Skewed

- Mean < Median **Conclusion:**

Median is better for skewed datasets because:

- It is not affected by outliers
- Gives a more representative central value

Mean is better when data is symmetric

3. Best Sampling Strategy (Campus Survey)

Problem

You want to survey students in a campus.

Possible Sampling Methods

1. Simple Random Sampling

- Randomly pick students
May miss important groups

2. Systematic Sampling

- Every 10th student Can introduce bias

3. Cluster Sampling

- Select entire classes Not diverse enough

4. Stratified Sampling (BEST)

Divide population into groups:

- Department (CS, IT, Mechanical)
- Year (1st, 2nd, 3rd, 4th)
- Then sample from each group.

Conclusion:

Stratified sampling is the best method for a campus survey because it ensures representation from different student groups such as departments and academic years, leading to more accurate and unbiased results compared to simple random or cluster sampling.